AUTOMATIC STREET LIGHT USING LDR

Project Report — Electronic Devices and Circuits (ECE)

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1. Abstract

This project presents a simple automatic street-light circuit using a light-dependent resistor (LDR) and an NPN transistor. The LDR senses the surrounding light level. During daytime, the circuit keeps the LED OFF; when the surroundings become dark, the transistor switches ON and the LED glows. The circuit can be assembled and tested using Tinkercad Circuits.

2. Aim

To design and simulate an automatic street-light circuit that switches an LED ON in darkness and OFF in bright light using an LDR and a transistor.

3. Objectives

• Understand the working of an LDR.

• Use an NPN transistor as an electronic switch.

• Simulate and observe the circuit under bright and dark conditions.

• Understand a basic application of sensor-based automation.

4. Components Required

Component	Purpose
LDR	Senses ambient light
NPN transistor (e.g., 2N2222)	Acts as an electronic switch
10 kΩ resistor	Provides the sensor/base bias path
220 Ω resistor	Limits LED current
LED	Represents the street light
5 V DC supply and wires	Power and connections

5. Circuit Connections

• Connect the +5 V supply to one end of the 10 kΩ resistor. Connect the other end to the BASE node.

• Connect the LDR between the BASE node and GND.

• Connect the transistor EMITTER to GND.

• Connect +5 V to a 220 Ω resistor, then to the LED anode. Connect the LED cathode to the transistor COLLECTOR.

• Connect all ground points to the same ground rail.

Important: Check the pinout of the transistor model in Tinkercad before wiring. The physical pin order can differ between transistor types.

6. Working Principle

An LDR is a resistor whose resistance changes with light intensity. Its resistance is low in bright light and high in darkness. In bright light, the LDR pulls the base node toward ground, keeping the transistor OFF; therefore, the LED does not glow. In darkness, the LDR resistance increases, allowing the base node voltage to rise through the 10 kΩ resistor. The transistor turns ON and current flows through the LED.

7. Simulation Procedure in Tinkercad

• Open https://www.tinkercad.com/ and sign in.

• Open Circuits and choose Create new Circuit.

• Add an LDR, NPN transistor, LED, 10 kΩ resistor, 220 Ω resistor and a 5 V supply.

• Connect the components as listed in Section 5. Verify the transistor terminals.

• Start Simulation and adjust the LDR light level if the simulator control is available.

• Observe the LED in bright and dark conditions and insert screenshots in the spaces below.

8. Observation Table

Condition	Expected transistor state	Expected LED state
Bright light	OFF	OFF
Darkness	ON	ON

Record the actual output from your simulation. If the LED does not switch as expected, recheck the LDR connection, common ground, transistor pinout and LED polarity.

9. Applications

• Automatic street and pathway lights.

• Garden and parking-area lighting.

• Night lamps and light-sensitive indicators.

10. Advantages and Limitations

Advantages: Simple circuit, low cost, automatic operation and easy simulation.

Limitations: Shadows, dust and nearby lights can affect sensing. A real outdoor street-light system may require a suitable driver or relay, a higher-power lamp and weatherproof protection.

11. Result

The proposed circuit is expected to switch the LED OFF in bright light and ON in darkness. The simulation demonstrates the light-sensing property of an LDR and the switching action of an NPN transistor.

12. Conclusion

The automatic street-light project demonstrates a basic application of electronic devices in daily life. The LDR senses the light level and the transistor controls the LED. The project is suitable for learning sensor-based switching and can be expanded to control a larger lamp using an appropriate driver circuit.

13. References

• Tinkercad Circuits: https://www.tinkercad.com/things/916dyfHY5Jm-automatic-street-light-using-ldr

• Electronic Devices and Circuits course notes — LDR characteristics and transistor switching.

• Datasheet of the selected NPN transistor and LDR, as applicable.